Week 9: Accessibility Audit Report

Web Programming - Memory Match: Neon Arcade
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Project URL: https://abinaah2004-maker.github.io/WebClass/teamproject/index.html

PURPOSE
This report documents the accessibility audit of the Memory Match game, explains why the initial results were lower, and records improvements made after reviewing Lighthouse findings. Each topic is covered once to avoid duplicated report sections.

1. INITIAL AUDIT RESULTS
Performance: 88/100 (a later screenshot also showed 77). Loading and rendering work remained; scores can vary with device, network, cache, and test conditions.
Accessibility: 96/100. Some automated accessibility checks still needed attention.
Best Practices: 77/100. Lighthouse reported warnings; inspect diagnostics before assigning a cause.
SEO: 90/100. The page was missing basic search metadata, including a meta description.

2. WHY THE SCORES WERE LOW
Performance: results indicated time spent rendering main content, possible render-blocking CSS, and long main-thread tasks. These can delay Largest Contentful Paint (LCP). Scores may differ between runs.
Accessibility: a score of 96 means at least one automated check was not passing. Manual checks include keyboard focus, logical tab order, visible focus, and focus handling for new content. These must be tested by a person.
Best Practices: the score of 77 showed warnings. The score alone does not identify the cause, so review diagnostics rather than guessing.
SEO: the score of 90 was associated with missing page metadata, including a description.

3. IMPROVEMENTS MADE
Added a skip link as the first useful keyboard navigation option so keyboard users can jump directly to the main content.
Added a main-content target using <main id="main-content" tabindex="-1"> so the skip link can move focus to the main area.

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4. IMPROVEMENTS MADE (CONTINUED)

Reviewed page structure and interactive controls with keyboard use in mind. Lighthouse manual checks still need hands-on verification.

Identified the SEO metadata issue for correction by adding a meaningful meta description.

Reviewed performance diagnostics, including render delay, render-blocking styles, caching opportunities, and long tasks. These are improvement targets; the report does not claim that every performance warning has been eliminated.

5. RESULTS AFTER IMPROVEMENT

A later Lighthouse screenshot shows Accessibility 100/100, an improvement from the earlier 96/100 result.

Another screenshot showed Performance 88, Accessibility 96, Best Practices 77, and SEO 90, demonstrating that results differed between audit runs. The report records both snapshots instead of presenting them as one test.

The 100/100 automated accessibility score is a strong result, but it does not guarantee that every user can use the site. Manual testing remains important: keyboard-only navigation, visible focus, mobile layout, labels, and status messages.

6. MANUAL TESTING CHECKLIST

Keyboard navigation: Use Tab and Shift+Tab to reach every control in a sensible order.

Skip link: Press Tab at page load, activate the skip link, and confirm focus moves to main.

Focus visibility: Confirm focused buttons, links, and cards have a visible focus indicator.

Game controls: Check difficulty, mode, cards, and restart controls for understandable names and states.

Small screens: Check narrow widths and browser zoom without horizontal scrolling where avoidable.

7. CONCLUSION

The audit identified opportunities to improve navigation, structure, performance, and search metadata. The clearest documented accessibility change was the skip link and main-content focus target; a later screenshot recorded a perfect automated accessibility score. The report presents the audit once and explains findings without repetition.